IETF 127 Hackathon

CoAP over BPv7

Direct CoAP Communication for Delay-Tolerant Space Networks

Implementation of `draft-ietf-core-coap-bp-05`

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Motivation

- Enable efficient and interoperable CoAP communication in space networks with high delay and disruptions.
- Demonstrate the latest CoAP over BPv7 specification **(draft-ietf-core-coap-bp-05).**

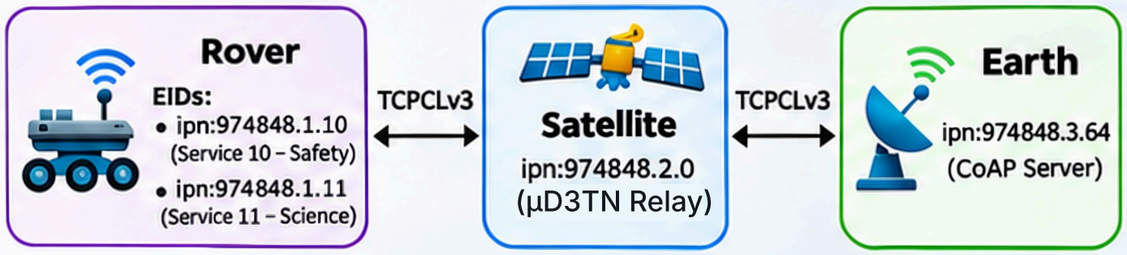
Key Contributions

- Direct CoAP over BPv7 (no proxy).
- Support for Single and Aggregate messages
- Three-component IPN EIDs (RFC 9758)
- Open, reproducible implementation, packet analysis, and tools

Experimental Setup

Platform	µD3TN + CoAP applications
Transport	TCPCLv3 (Earth analysis port: 4773)
Analyzer	Wireshark 4.6.9 (with Lua dissector)
IPN Allocator	974848
Services	• 11 – Rover Science • 10 – Rover Safety • 64 – Earth CoAP

System Architecture



Experiment Results

Single Bundle (Safety Guidance)

- CoAP Method : GET
- BP payload : 59 bytes
- Bundle lifetime : 100,000 ms (logs)
- CoAP response : 2.05 Content

Wireshark 4.6.9 — Single CoAP-over-BP

```
▼ DTN Bundle Protocol Version 7
    Payload Size: 59
▼ CoAP over BPv7
  ▼ CoAP Single message #1 - 0.01 (GET)
      Source: ipn:974848.1.10
      Destination: ipn:974848.3.64
      MID: 10734772; Token: 684d4727
```

Aggregate Bundle (Science Observations)

- CoAP Methods : POST + POST
- BP payload : 212 bytes
- Bundle lifetime : 90,000 ms (100 s – 10 s)
- CoAP responses : two × 2.01 Created
- Payload-length : 50 B each (opt. 65000)

Wireshark 4.6.9 — Aggregate CoAP-over-BP

```
▼ DTN Bundle Protocol Version 7
    Lifetime: 90000ms
▼ CoAP over BPv7
  ▼ Aggregate, 2 CoAP message(s)
    ▼ CoAP message #1 - 0.02 (POST)
        Source: ipn:974848.1.11
        Destination: ipn:974848.3.64
        MID: 2543436; Token: 2f8ecb82
    ▼ CoAP message #2 - 0.02 (POST)
        Source: ipn:974848.1.11
        Destination: ipn:974848.3.64
        MID: 2543437; Token: 715deaac
```

Wireshark 4.6.9: IPN Encoding (RFC 9758)

- ▼ **Destination Endpoint ID: ipn:974848.3.64**
  - Scheme Code: IPN (2)
  - IPN Element Count: 3
  - IPN Allocator: 974848
  - IPN Node Number: 3
  - IPN Service Number: 64
- ▼ **Source Node ID: ipn:974848.1.11**
  - Scheme Code: IPN (2)
  - IPN Element Count: 3
  - IPN Allocator: 974848
  - IPN Node Number: 1
  - IPN Service Number: 11

Results Summary

Interaction	Result
Safety GET (Single)	✓ 4 / 4 (2.05 Content)
Science POST (Single)	✓ 1 / 1 (2.01 Created)
Science POST (Aggregate)	✓ 2 / 2 (2.01 Created)
Token Matching	✓ All matched
BPv7 Lifetime	✓ Verified in packet (90,000 ms, Aggregate)
IPN Encoding	✓ Three-element (RFC 9758)

Takeaways

- ✓ Functional CoAP over BPv7 implementation (Draft-05)
- ✓ Successful Single and Aggregate operation
- ✓ Verified BPv7 lifetime and token matching
- ! **Experimental:** NSTART = 2 and Payload-length opt. 65000
- ! **Not yet tested** under long outages or congestion

GitHub Repository

ENTEL-WNG/CoAPoverBP-IPN-Draft05-NoProxy

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